

Interpretable Crisis Behavior Analysis Using Mobility and Social Media Data

Muhammad Hamza Arshad Majeed, Sidahmed Benabderrahmane, Talal Rahwan

New York University (NYUAD), Division of Science, Computer Science Department
mm12283@nyu.edu

Abstract. Crises alter both how people move and how they communicate. During emergencies such as wildfires and pandemics, changes in mobility patterns and online emotional discourse evolve jointly, yet they are typically studied in isolation. This paper presents a unified and interpretable pipeline that integrates mobility and social media data to identify cross-domain behavioral patterns that emerge in crisis settings. The framework is evaluated through two case studies: a short-term analysis of the January 2025 Los Angeles wildfires (prototype case, 33 days) and a long-term study of behavioral shifts during the COVID-19 pandemic in the UAE from March 2020 to December 2021 (primary case, 671 days). The pipeline aligns heterogeneous daily signals, transforms them into binary behavioral states, and applies Formal Concept Analysis (FCA) to extract co-occurrence structures. It then mines association rules and validates their stability through chronological holdout testing (for the longitudinal COVID case). A structured policy-translation layer provides robust rules as operational briefs specifying triggers, lead times, and action playbooks. The results reveal a clear cross-domain behavioral structure in both crises. In the wildfire case, traffic congestion, fear/anger sentiment, and governance discourse are tightly coupled within the 33-day window, with cross-domain rules reaching lift scores up to 2.5. In the COVID case, frequent mobility shifts and emotional volatility yielded 7 stable same-day rules (87% holdout pass rate) and 16 clean predictive rules with 2–7 day lead horizons. The work demonstrates that combining social media and mobility data can deliver crisis insights that are both scientifically rigorous and directly usable by policymakers.

Availability: <https://github.com/HamzaArshad2004/css-project-wildfire>,
<https://github.com/HamzaArshad2004/css-project-covid>

Keywords: Crisis Informatics · Formal Concept Analysis · Mobility Analysis · Social Media Analysis · Interpretable Machine Learning

1 Introduction

Human behavior during crises is visible across both physical and digital spaces. People alter where they go, how frequently they travel, and how they express risk, uncertainty, trust, and frustration online. Studying only one of these channels yields an incomplete account of human response, and potentially misleads decision-makers who must act under uncertainty. This paper is motivated by the premise

that crisis behavior should be modeled jointly across mobility and discourse. The practical rationale is direct: decision-makers need to know not only what the public is feeling, but whether those feelings accompany real behavioral shifts. That distinction matters for timing communication, targeting intervention, and overall operational readiness. The paper addresses two qualitatively different crises to test whether behavioral coupling is a general phenomenon or an artifact of a single event type. The January 2025 Los Angeles wildfires represent an acute, geographically concentrated, rapidly evolving emergency. The UAE COVID-19 pandemic (2020–2021) represents a prolonged, policy-mediated, multi-wave disruption spanning 671 days and four distinct policy phases: initial lockdown (Mar–May 2020), phased reopening (Jun–Dec 2020), mass vaccination rollout (Jan–Aug 2021), and endemic transition (Sep–Dec 2021). Together, the two cases test whether the framework transfers across acute and prolonged crisis settings.

Research Questions: This work addresses four central questions:

1. How do mobility and online discourse signals co-evolve during crises of different types and durations?
2. Which cross-domain behavioral patterns recur reliably within each crisis, and can they survive conservative temporal validation?
3. Which patterns are contemporaneous, and which provide meaningful predictive lead time suitable for anticipatory intervention?
4. Do behavioral patterns transfer across acute and prolonged crisis types?

Contributions: The paper makes four primary contributions:

- An interpretable mobility–discourse fusion pipeline grounded in binary behavioral states and Formal Concept Analysis;
- A two-case evaluation covering a 33-day LA wildfire case (Caltrans PeMS and Reddit) and a 671-day UAE COVID-19 case (Google Mobility and Reddit);
- A reliability-first rule framework combining multi-stage pruning and chronological holdout testing;
- A structured policy-brief format translating validated rules into agency-ready operational guidance with named bodies, lead times, and action playbooks.

2 Related Work

Crisis mobility. Prior work has shown that large-scale crises strongly alter human movement. Beria and Lunkar [2] used anonymized location-based services data to study mobility restrictions during the first COVID-19 wave in Italy, while Caselli et al. [3] showed that post-lockdown mobility recovery was shaped by policy and socioeconomic factors and could be partly forecast from early crisis data. These studies establish mobility as a key behavioral signal, but they largely treat movement as a standalone outcome rather than modeling its relationship with online discourse. This paper addresses this gap by jointly analyzing mobility and discourse at daily resolution.

Social media as a behavioral sensor. Social media has also been widely used as a real-time sensor of public response during emergencies. Imran et al. [6] surveyed machine learning pipelines for crisis-related social media processing, Zhang et al. [7] reviewed disaster-oriented public information systems, and Zubiaga et al. [8] studied rumor detection in noisy, fast-evolving crisis content. For sentiment quantification, this paper uses VADER [5], a rule-based model designed for social media text and suitable when labeled crisis-specific data are unavailable. However, these works typically analyze online discourse separately from physical mobility.

Integration gap. Prior studies establish mobility and social discourse as valuable but mostly separate crisis signals. Existing multi-modal approaches often favor predictive accuracy through latent embeddings or black-box classifiers, limiting auditability for emergency practitioners. This paper addresses this gap with a transparent rule-based fusion framework that links mobility, sentiment, emotion, and topic states, validates the resulting patterns temporally, and translates robust rules into operational guidance.

3 Data and Method

Case studies and data sources. We evaluate the framework on two crises. The prototype case examines January 2025 wildfire behavior in Los Angeles over a 33-day window (January 3–February 4, 2025), covering four phases: pre-ignition baseline (Jan 3–6), rapid ignition and spread of the Palisades and Eaton fires (Jan 7–11), peak crisis with approximately 180,000 residents evacuated (Jan 12–20), and progressive containment and return orders (Jan 21–Feb 4). Mobility signals are daily Caltrans PeMS traffic metrics: Vehicle Miles Traveled (VMT), Vehicle Hours Traveled (VHT), Travel Time Index (TTI), and vehicle delay. Social discourse is collected with PRAW Python API from `r/LosAngeles` and `r/wildfires`, then processed with VADER sentiment scoring and keyword-based emotion detection. The primary case covers UAE COVID-19 behavior from March 1, 2020 to December 31, 2021 (671 days), combining national-level Google Community Mobility Reports with COVID-related discourse from `r/UAE`, `r/dubai`, `r/abudhabi`, `r/Coronavirus`, and `r/worldnews`. The UAE timeline is divided into four policy phases: initial lockdown (Mar–May 2020), phased reopening (Jun–Dec 2020), vaccination rollout (Jan–Aug 2021), and endemic transition (Sep–Dec 2021). In both cases, mobility and discourse signals are synchronized to a daily index; Reddit posts are deduplicated by post ID, and days with fewer than 5 posts are flagged as low-coverage.

Behavioral-state encoding. The core design choice is to represent crisis dynamics as interpretable binary behavioral states. Each observation day is encoded as a set of semantically meaningful attributes, enabling transparent rule extraction, hierarchical concept analysis, and policy-legible interpretation. Feature engineering produces a 19-column LA matrix and a 29-column UAE matrix covering

five groups of states: *mobility states*, such as retail/workplace/transit drops, residential increase, grocery spike, and traffic congestion; *sentiment states*, such as high negative sentiment, high positive sentiment, and day-over-day sentiment shift; *emotion/topic states*, such as fear, anger, sadness, solidarity, health concern, vaccine mention, and *policy/governance discussion*; composite states, such as `emotion_with_mobility_signal`, `emotion_mobility_mismatch`, and `calm_mobile_baseline`. Crucially, `emotion_mobility_mismatch` flags days with active online emotion/topic features but no physical mobility or traffic anomalies, capturing sustained crisis discourse or collective sense-making that continues even when physical movement returns to baseline. *Temporal states* use lagged antecedents at 1, 3, and 7 days and lead outcomes at 2 and 7 days. Continuous signals are binarized using documented thresholds, including VADER polarity cutoffs, mobility deviations from baseline, and rolling Caltrans congestion baselines; full threshold definitions are provided in the Github repository. UAE thresholds are relative to the Google Mobility pre-pandemic baseline, while LA thresholds use a 30-day rolling pre-fire Caltrans average. The feature `dominant_emotion_fear` is active when fear keywords exceed all other emotion categories on a given day.

Interpretable FCA and rule-mining layer. The interpretable modeling layer is based on Formal Concept Analysis (FCA) [4] and association rule mining [1]. Each crisis day is represented as a formal context, where objects are observation days and attributes are binary mobility, sentiment, emotion, topic, composite, or temporal states. FCA organizes this binary context into a concept lattice, where each concept corresponds to a set of days sharing a set of behavioral attributes. To make this step concrete, Fig. 1 gives a small wildfire-like example using the same type of daily crisis features as our data: traffic congestion, anger, fear, policy discourse, sadness, mismatch, and calm baseline. The figure shows how daily binary observations are transformed into a formal context, how selected shared patterns appear in a lattice excerpt, and how association rules summarize interpretable behavioral regularities. This makes the resulting patterns auditable: for example, days combining traffic congestion and anger can be directly inspected and linked to fear, policy discourse, or mobility disruption. Temporal lag/lead features extend this layer from same-day pattern discovery to short-term prediction, while chronological validation checks whether the rules persist beyond the phase in which they were discovered.

FCA construction and rule extraction. For the UAE case, objects are aggregated by weekly majority vote for lattice construction to avoid combinatorial explosion (strictly a visual/interpretative tool to render an auditable lattice snippet), while rule mining runs on the full 671-day daily data. Same-day association rules are extracted using fixed thresholds: UAE rules require support $\geq 10\%$, confidence $\geq 75\%$, and lift ≥ 1.8 ; LA rules require support $\geq 10\%$, confidence $\geq 80\%$, lift ≥ 1.05 , maximum premise size = 2, and maximum conclusion prevalence $\leq 75\%$. A tautology filter removes rules whose conclusions follow from composite-feature premises by definition, and rules are tagged `cross_domain = True` when they span both mobility and emotion/discourse feature sets.

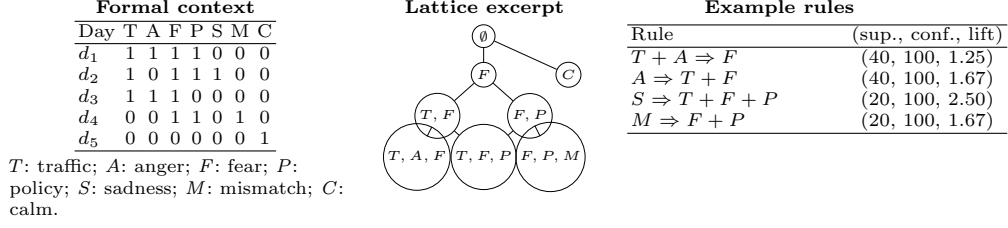


Fig. 1: Illustrative FCA example for crisis behavior analysis. The formal context encodes daily mobility–discourse states; the lattice excerpt shows selected shared behavioral patterns; and the rules summarize interpretable co-occurrences between traffic, emotion, policy discourse, mismatch, and calm states. Values denote support, confidence, and lift.

Temporal prediction, pruning, and validation. In the UAE case, in order to identify predictive structure, lag and lead features are generated by shifting signals across 1-, 3-, and 7-day antecedent windows and 2- and 7-day consequent windows, yielding a 657×117 UAE feature matrix (14 undefined boundary rows are dropped). Both same-day and predictive rules pass through a shared pruning pipeline applied after the statistical thresholds. The temporal stages described here operate only on lag/lead features, so same-day rules pass through them unchanged; same-day rules are therefore reduced by the redundancy stages alone (tautology, bidirectional, and subsumption pruning), from 23 to 8 in the UAE case. For predictive candidates, the pipeline first applies a temporal direction check that discards rules whose consequent carries a lag suffix, ensuring conclusions do not predate premises, and simultaneously removes any unsuffixed (contemporaneous) rules from the predictive set to prevent duplication. Next, a leakage filter drops lead features from antecedents. A persistence filter then eliminates within-variable loops where a feature’s base name appears in both components (e.g., a `grocery_spike` predicting a later `grocery_spike`). Finally, the redundancy pass clears tautological feature groups, bidirectional duplicates, and rules subsumed by stronger antecedents, capped at three rules per consequent, reducing the raw predictive candidates to 16 clean rules. A 70/30 chronological split validates the UAE data (training: Mar 2020–Jun 2021; holdout: Jul 2021–Dec 2021), filtering out rules with a holdout confidence drop greater than 10 percentage points. The LLM-assisted step is not used to validate, rank, or select rules; statistical filtering, multi-stage pruning, and chronological backtesting alone determine which rules are retained. After this pipeline, GPT-4o-mini is applied purely as a qualitative annotation layer, translating each validated rule into policy-facing language, a plain-language rationale and a named-agency recommendation. Prompts integrate the historical crisis timeline and mandate localized, named-agency action steps (e.g., NCEMA for UAE; LAFD for LA). The LA wildfire prototype mines same-day rules only and is therefore evaluated by the statistical thresholds and the inline tautology filter alone; with no lag/lead features and a compressed 33-day window, it requires neither the temporal-pruning stages nor chronological backtesting.

Policy-brief generation. Each validated rule is translated into a structured operational brief specifying trigger conditions, expected outcome, statistical strength (support, confidence, lift, conviction, leverage), signal window, agency action playbook, false-alarm risk level, and confirmation guidance. Predictive briefs additionally report `lead_time_days` (pre-computed from the rule structure) and a named responsible agency.

4 Results

4.1 Case Study I: LA Wildfires

Behavioral activation profile. Over the 33-day wildfire window (19 features), the most active binary states were: solidarity messages (54.5%), policy/governance discussion (48.5%), traffic congestion (42.4%), fear keywords (39.4%), sentiment shift (39.4%), high negative sentiment (33.3%), and sadness keywords (33.3%).

Top cross-domain wildfire rules. Five representative high-value cross-domain rules are summarized in Table 1; the three most important are described below.

Table 1: Top Cross-Domain Association Rules: LA Wildfires

Antecedent $\rightarrow$ Consequent	Supp.	Conf.	Lift
traffic congestion + anger $\rightarrow$ fear keywords	15.2%	100%	2.54
traffic congestion + sadness $\rightarrow$ policy/governance	18.2%	83.3%	1.72
traffic congestion + anger $\rightarrow$ policy/governance	15.2%	80.0%	1.65
fear + anger $\rightarrow$ traffic congestion	18.2%	83.3%	1.96
anger + policy/governance $\rightarrow$ traffic congestion	15.2%	80.0%	1.89

Some of these rules reach 100% confidence; given the 33-day window, they rest on 4–6 supporting days and are best read as strong within-prototype signals to be confirmed on larger samples rather than population-level certainties.

W1: `traffic_congestion+anger` $\Rightarrow$ `fear_keywords`. On every day that combined traffic congestion with anger discourse, fear keywords also appeared. This rule suggests that physical immobility and anger toward conditions reliably co-occur with the escalation to explicit fear expression, a pattern consistent with acute, place-based threat.

W2: `traffic_congestion+sadness` $\Rightarrow$ `policy/governance`. On days where congestion co-occurred with grief and loss discourse, public accountability framing consistently emerged. Effective public communication is most urgently needed precisely when communities experience both physical disruption and emotional loss simultaneously.

W3: `fear+anger` $\Rightarrow$ `traffic congestion`. The reverse coupling reached 83.3% confidence and lift 1.96, indicating that a fear–anger climate was nearly twice as likely as chance to co-occur with detectable traffic congestion. This is the most operationally relevant direction: discourse signals offer a potential warning indicator for congestion-management agencies.

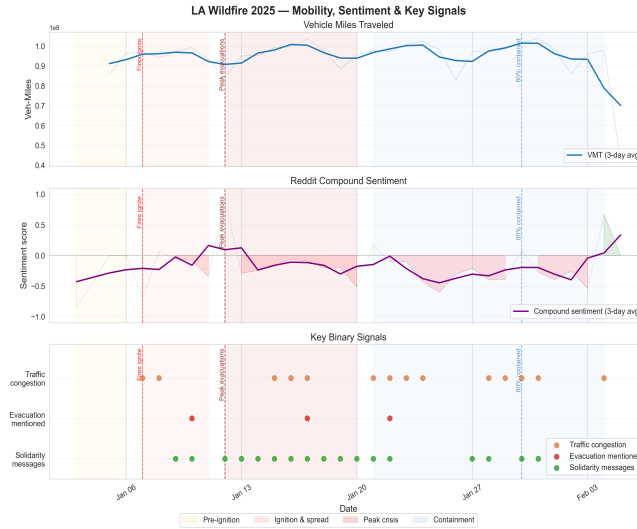


Fig. 2: LA wildfire analysis: co-evolution of Caltrans mobility indicators and Reddit discourse during the Jan–Feb 2025 wildfire event, with crisis-phase shading.

4.2 Case Study II: UAE COVID-19

Behavioral activation profile. Over the 671-day COVID window (29 features), major binary-state activation rates were:

- covid_topic_detected, dominant_emotion_fear: 84.9%
- sentiment_shift_detected: 58.4%; health_concern: 58.3%
- mobility_drop_transit: 54.1%; vaccine_mentioned: 45.0%
- solidarity_messages: 44.0%; residential_increase: 43.1%
- grocery_spike: 42.6%; calm_mobile_baseline: 39.6%

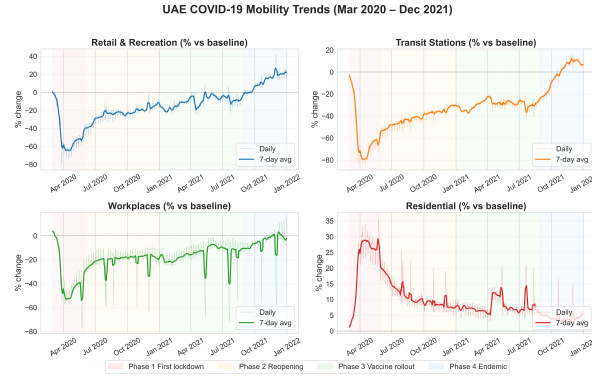
While the high baseline for the COVID topic is expected since the data comes from targeted subreddits, the true signal lies in how specific emotions and mobility shifts fluctuate around this baseline as the pandemic progressed. The macro-level temporal dynamics of mobility and discourse are shown in Figure 3.

Rule quality outcomes. The pipeline applied conservative filtering: 23 raw same-day rules extracted; 8 after pruning; 7 stable after chronological holdout (87%); and 16 clean predictive rules after pruning. The one rule that failed, `grocery_spike + weekend` $\rightarrow$ `calm_mobile_baseline` showed a 12.2 pp confidence drop on the holdout period, consistent with the fading of pandemic-specific weekend routines in Phase 4.

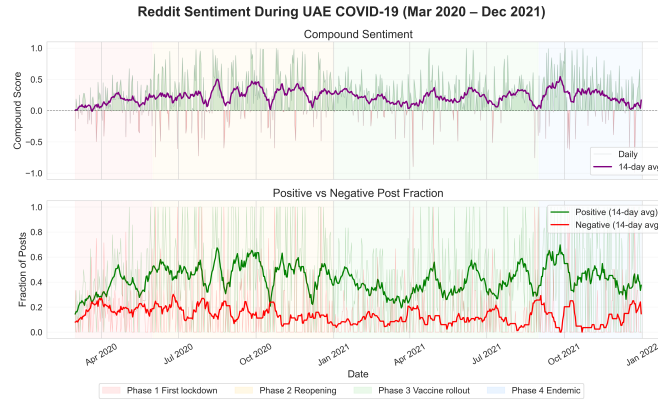
Top stable same-day COVID rules:

C1: `health_concern + calm_mobile_baseline` $\Rightarrow$ `grocery_spike` (sup.=20.4%, conf.=77.4%, lift=1.82)

C2: `grocery_spike + high_positive_sentiment` $\Rightarrow$ `calm_mobile_baseline` (sup.=15.4%, conf.=85.4%, lift=2.16)



(a) UAE mobility trends with 7-day rolling average and policy-phase shading.



(b) VADER compound score and positive/negative fractions with 14-day rolling average.

Fig. 3: Macro temporal dynamics of movement and discourse during UAE COVID-19. Sharp mobility drops in Phase 1 and partial recovery in Phases 3–4 are visible alongside the persistent fear-dominant sentiment baseline.

C3: `grocery_spike+vaccine_mentioned` $\Rightarrow$ `calm_mobile_baseline` (sup.=26.2%, conf.=79.5%, lift=2.01)

Together, these rules describe the transition from latent health concern to behavioral normalization. C1 indicates that grocery-demand surges can emerge even on otherwise routine mobility days when health concerns remain elevated. In contrast, C2 and C3 associate grocery activity with positive sentiment and vaccine discussion, suggesting that later-period retail activity increasingly reflected adaptation and recovery rather than anxiety-driven stockpiling. The persistence of C3 across 26% of all days highlights the stability of this normalization phase.

Top predictive COVID rules (lag/lead):

P1: `calm_mobile_baseline + high_positive_sentiment_lag1` $\Rightarrow$ `grocery_spike_lead7` (7-day lead) (sup.=14.6%, conf.=91.7%, lift=2.11)

P2: `calm_mobile_baseline + sentiment_worsened_lag7` $\Rightarrow$ `grocery_spike_lead7` (7-day lead) (sup.=11.1%, conf.=91.8%, lift=2.11)

P3: `lockdown_mentioned + lockdown_mentioned.lag3 ⇒ residential.increase_lead2` (2-day lead) (sup.=12.0%, conf.=79.7%, lift=1.81)

Together these rules show that sentiment and discourse forecast distinct behavioral outcomes at operationally useful horizons. A calm mobility baseline combined with the prior day’s positive sentiment anticipates a grocery spike a full week ahead (P1); notably, worsened sentiment seven days earlier forecasts the same outcome with near-identical strength (P2), indicating that grocery surges follow sentiment volatility in either direction rather than positive affect alone. At a shorter horizon, persistent lockdown discourse, mentioned both today and three days prior, forecasts a measurable shift toward residential confinement within 48 hours (P3), a separate mobility outcome useful for anticipating compliance. Values denote support, confidence, and lift.

4.3 Cross-Case Comparison

Table 2 summarizes the structural differences between the two case studies. Together, both cases confirm that coupling between discourse and mobility is real but conditional on crisis type, phase, and feature combination. Across both crises, cross-domain rules cluster at higher lift values than intra-domain rules, indicating that mobility–discourse coupling carries more statistical information than within-domain associations alone. The `emotion.mobility_mismatch` state further illustrates the crisis-type dependence: it fires on only 5.8% of COVID days, against 27.3% in the wildfire case, since an acute emergency leaves concern lingering after movement normalizes whereas a prolonged pandemic suppressed both in tandem.

Table 2: Comparison of Wildfire and COVID Case Study Outcomes

Dimension	LA Wildfire	UAE COVID-19
Observation days	33	671
Binary features	19	29
Raw same-day rules	63 (passed thresholds)	23
After pruning	47 (after tautology)	8
Holdout reliability	Prototype only	87% (7/8)
Predictive rules	Limited	16 clean
Dominant discourse	Governance + solidarity	COVID salience + fear
Dominant mobility	Traffic congestion	Transit suppression + grocery cycles
Mobility source	Caltrans PeMS	Google Mobility Reports
Discourse source	r/LosAngeles	Reddit UAE/COVID subreddits
Top conf./lift	100% / 2.54 (W1)	91.8% / 2.11 (P2)
Primary rule value	Acute coupling	Stable + predictive temporal structure

5 Discussion and Policy Implications

Policy translation. Each validated rule is converted into an operational brief specifying trigger conditions, expected outcome, statistical strength (support, confidence, lift, conviction, leverage), signal window, agency-specific action playbook, and false-alarm guidance.

Policy insights. The rules reveal four actionable crisis-management signals. Firstly, hidden risk can appear under apparent calm: when `health.concern` rises

during `calm_mobile_baseline` periods, same-day grocery spikes become more likely (C1, conf. 77.4%), suggesting that NCEMA-style emergency authorities should pair reassurance messaging with supply checks before visible mobility disruption appears. Second, a calm mobility baseline combined with shifting sentiment, positive or negative, forecasts a grocery-demand spike a week ahead (P1–P2; conf. 91.7–91.8%), giving NCEMA and logistics planners a seven-day window to pre-position supply. Third, persistent lockdown discourse forecasts a shift toward residential confinement within 48 hours (P3, conf. 79.7%), letting authorities anticipate compliance ahead of visible mobility change. Fourth, in the wildfire case, traffic congestion and negative emotion co-occur with governance accountability discourse (W2–W3), suggesting that LAFD, CAL FIRE, and LA County OES should prioritize transparent updates when traffic congestion and emotional distress rise together.

6 Conclusion and Future Work

This paper introduced an interpretable framework for integrating mobility and social discourse signals to analyze crisis behavior. Across a 33-day LA wildfire prototype and a 671-day UAE COVID-19 case study, the framework extracted cross-domain behavioral structures, validated them through pruning and chronological holdout testing, and translated them into operational guidance. The results show that multi-modal behavioral rules provide same-day and short-term signals, including 87% holdout stability and 2–7 day predictive lead windows, supporting communication strategy, logistics readiness, and crisis-response planning. Future work will extend the framework to finer geographic resolution, adaptive dashboards, crisis-calibrated sentiment models, and broader transfer tests across earthquakes, heatwaves, political crises, and other emergencies.

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